

PRAKTIKUM SISTEM OPERASI

MODUL 13

Perintah Dasar Linux



Oleh:

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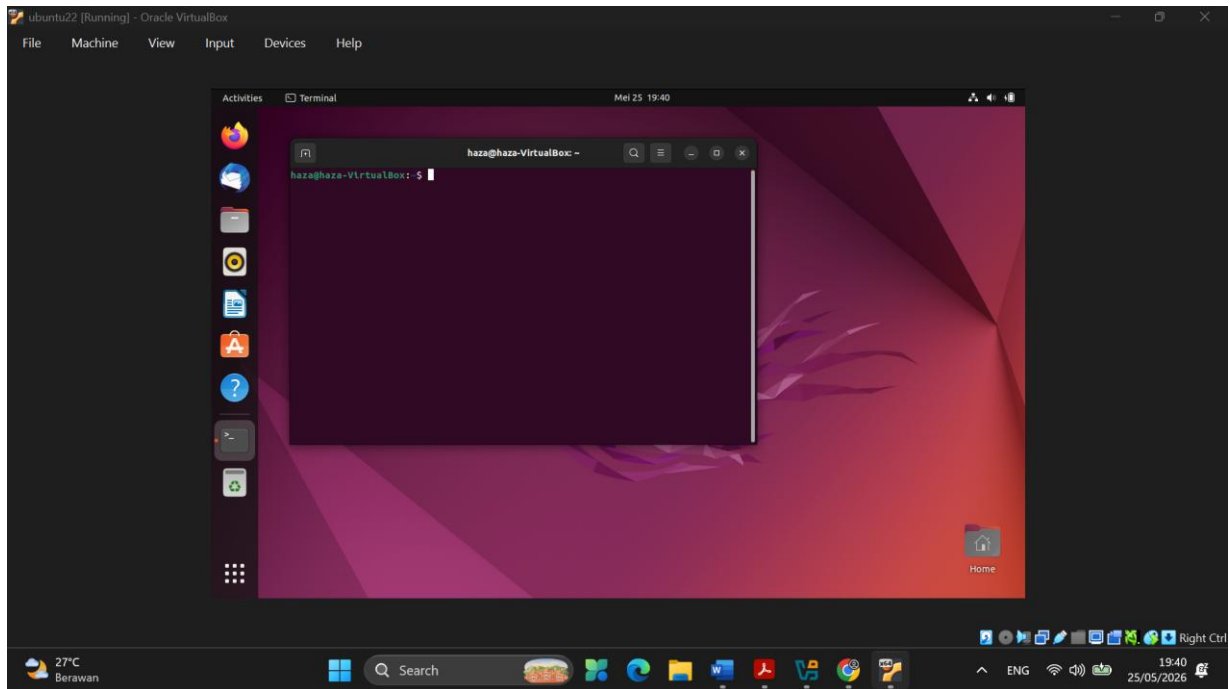
Program Studi S1 Rekayasa Perangkat Lunak

DIREKTORAT KAMPUS PURWOKERTO UNIVERSITAS TELKOM

2026

Jurnal

1.



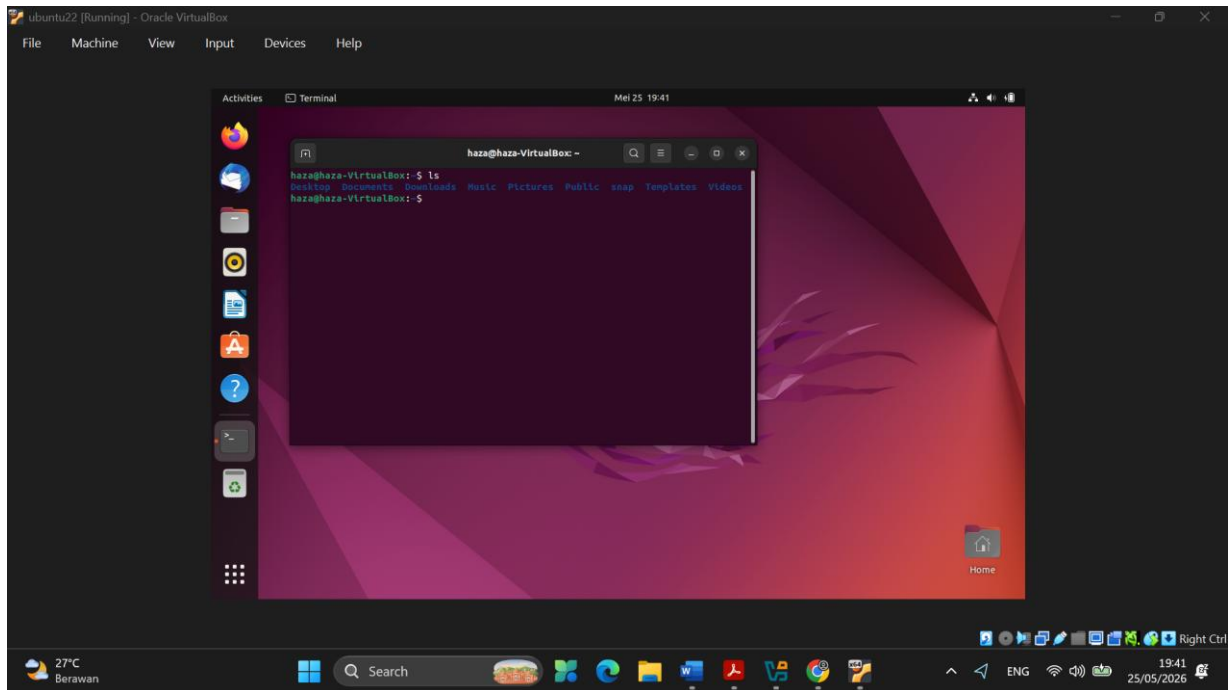
haza@haza-VirtualBox:~\$

Artinya:

- haza = username
- haza-VirtualBox = nama komputer/host
- ~ = home directory user
- \$ = user biasa (bukan root)

2.

a. ls



b. Option dan parameter dari perintah ls

Perintah:

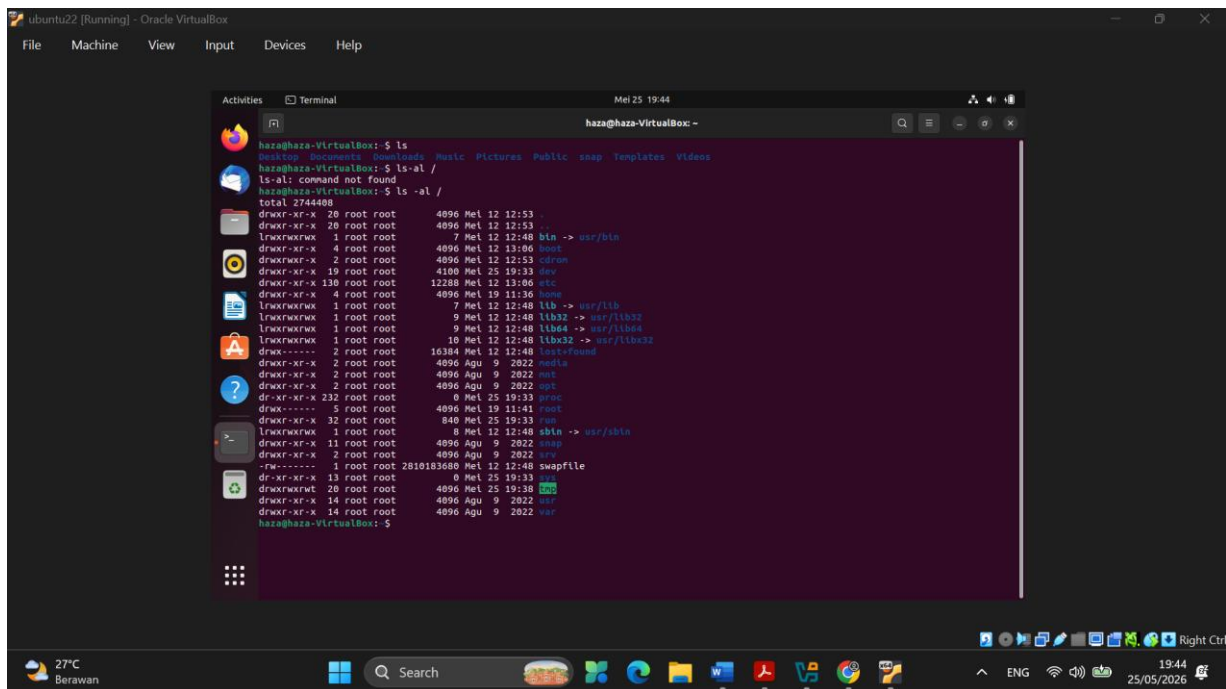
ls

- option = tidak ada
- parameter = tidak ada

c. Fungsi perintah ls

Fungsi perintah ls adalah untuk menampilkan isi direktori/folder saat ini.

d. `ls -al /`



```
haza@haza-VirtualBox:~$ ls
Desktop Documents Downloads Music Pictures Public snap templates Videos
haza@haza-VirtualBox:~$ ls -al /
ls: al: command not found
haza@haza-VirtualBox:~$ ls -al /
total 2744480
drwxr-xr-x 28 root root      4096 Mei 12 12:53 .
drwxr-xr-x 28 root root      4096 Mei 12 12:53 ..
lrwxrwxrwx 1 root root         7 Mei 12 12:48 bin -> usr/bin
drwxr-xr-x 4 root root      4096 Mei 12 11:06 boot
drwxrwxr-x 2 root root      4096 Mei 12 12:53 cdrom
drwxr-xr-x 19 root root     4100 Mei 25 19:33 dev
drwxr-xr-x 138 root root    12288 Mei 12 13:06 etc
drwxr-xr-x 4 root root      4096 Mei 19 11:36 home
lrwxrwxrwx 1 root root         7 Mei 12 12:48 lib -> usr/lib
lrwxrwxrwx 1 root root         9 Mei 12 12:48 lib32 -> usr/lib32
lrwxrwxrwx 1 root root         9 Mei 12 12:48 lib64 -> usr/lib64
lrwxrwxrwx 1 root root        10 Mei 12 12:48 libx32 -> usr/libx32
drwx----- 2 root root    16384 Mei 12 12:48 lost+found
drwxr-xr-x 2 root root      4096 Agu 9 2022 media
drwxr-xr-x 2 root root      4096 Agu 9 2022 mnt
drwxr-xr-x 2 root root      4096 Agu 9 2022 opt
dr-xr-xr-x 232 root root         0 Mei 25 19:33 proc
drwx----- 5 root root      4096 Mei 19 11:41 run
drwxr-xr-x 32 root root      840 Mei 25 19:33 run
lrwxrwxrwx 1 root root         8 Mei 12 12:48/sbin -> usr/sbin
drwxr-xr-x 11 root root      4096 Agu 9 2022 snap
drwxr-xr-x 2 root root      4096 Agu 9 2022 sys
-rw----- 1 root root 2810183680 Mei 12 12:48 swapfile
dr-xr-xr-x 13 root root         0 Mei 25 19:33 sys
drwxrwxrwt 26 root root      4096 Mei 25 19:38 tmp
drwxr-xr-x 14 root root      4096 Agu 9 2022 usr
drwxr-xr-x 14 root root      4096 Agu 9 2022 var
haza@haza-VirtualBox:~$
```

e. Option dan parameter dari perintah `ls -al /`

- option = `-al`
- parameter = `/`

Keterangan:

- `a` = menampilkan file tersembunyi
- `l` = menampilkan detail file

f. `ls -al /`

Menampilkan seluruh isi direktori root / secara lengkap/detail termasuk hidden file.

g. Mengapa hasil perintah `a` dan `d` berbeda?

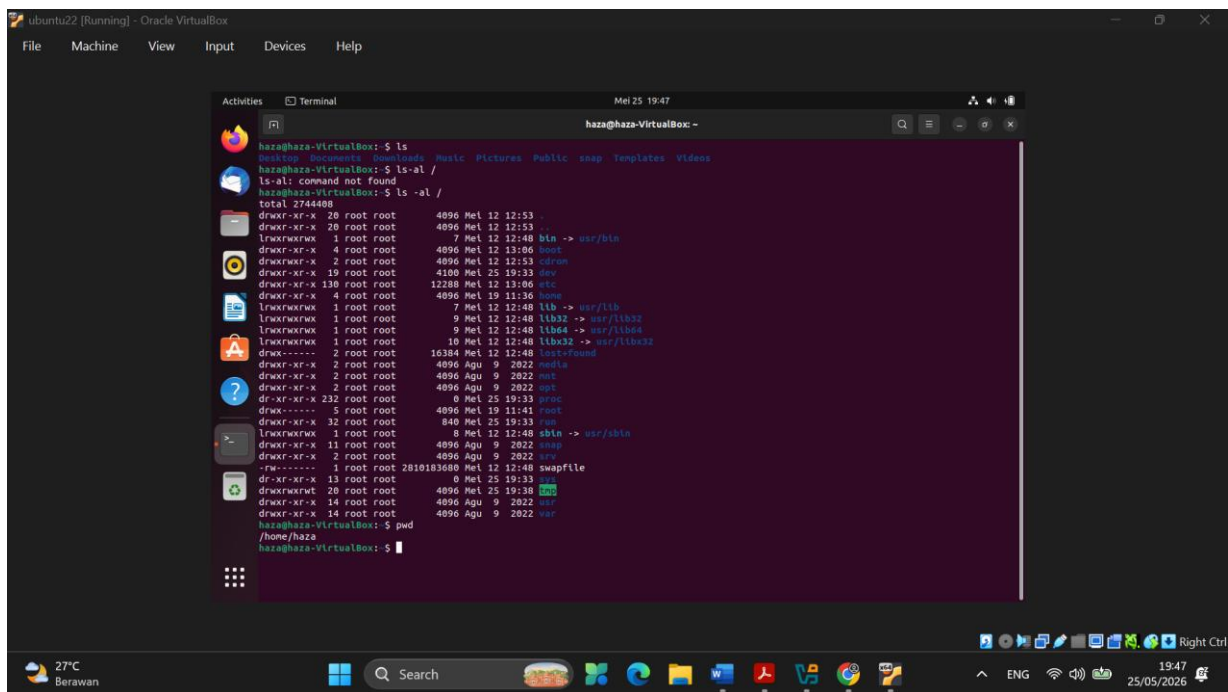
Karena:

- `ls` hanya menampilkan isi folder saat ini
- `ls -al /` menampilkan isi direktori root / secara detail lengkap beserta hidden file sehingga hasilnya lebih banyak dan berbeda.

3.

a. jalankan perintah

pwd



The screenshot shows a terminal window titled 'haza@haza-VirtualBox: ~' within an Oracle VM VirtualBox environment. The terminal displays the following commands and their outputs:

```
haza@haza-VirtualBox:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  snap  templates  Videos
haza@haza-VirtualBox:~$ ls -al /
ls: al: command not found
haza@haza-VirtualBox:~$ ls -al /
total 2744480
drwxr-xr-x 20 root root      4096 Met 12 12:53 .
drwxr-xr-x 20 root root      4096 Met 12 12:53 ..
lrwxrwxrwx 1 root root         7 Met 12 12:48 bin -> usr/bin
drwxr-xr-x 4 root root      4096 Met 12 11:06 boot
drwxrwxr-x 2 root root      4096 Met 12 12:53 cdrom
drwxr-xr-x 19 root root    4100 Met 25 19:33 dev
drwxr-xr-x 136 root root   12288 Met 12 11:06 etc
drwxr-xr-x 4 root root      4096 Met 19 11:36 home
lrwxrwxrwx 1 root root         7 Met 12 12:48 lib -> usr/lib
lrwxrwxrwx 1 root root         9 Met 12 12:48 lib32 -> usr/lib32
lrwxrwxrwx 1 root root        10 Met 12 12:48 lib64 -> usr/lib64
drwx----- 2 root root   16384 Met 12 12:48 lost+found
drwxr-xr-x 2 root root      4096 Agu 9 2022 media
drwxr-xr-x 2 root root      4096 Agu 9 2022 mnt
drwxr-xr-x 2 root root      4096 Agu 9 2022 opt
dr-xr-xr-x 232 root root      0 Met 25 19:33 proc
drwx----- 5 root root      4096 Met 19 11:41 root
drwxr-xr-x 32 root root      840 Met 25 19:33 run
lrwxrwxrwx 1 root root         8 Met 12 12:48/sbin -> usr/sbin
drwxr-xr-x 11 root root      4096 Agu 9 2022 snap
drwxr-xr-x 2 root root      4096 Agu 9 2022 srv
-rw----- 1 root root 2810183680 Met 12 12:48 swapfile
dr-xr-xr-x 13 root root      0 Met 25 19:33 sys
drwxr-xr-x 26 root root      4096 Met 25 19:30 tmp
drwxr-xr-x 14 root root      4096 Agu 9 2022 usr
drwxr-xr-x 14 root root      4096 Agu 9 2022 var
haza@haza-VirtualBox:~$ pwd
/home/haza
haza@haza-VirtualBox:~$
```

b. Option dan parameter dari perintah pwd

- option = tidak ada
- parameter = tidak ada

c. fungsi perintah pwd

Fungsi perintah `pwd` adalah untuk menampilkan lokasi/path direktori saat ini.

4.

a. Jalankan perintah

Ketik:

`cd /`

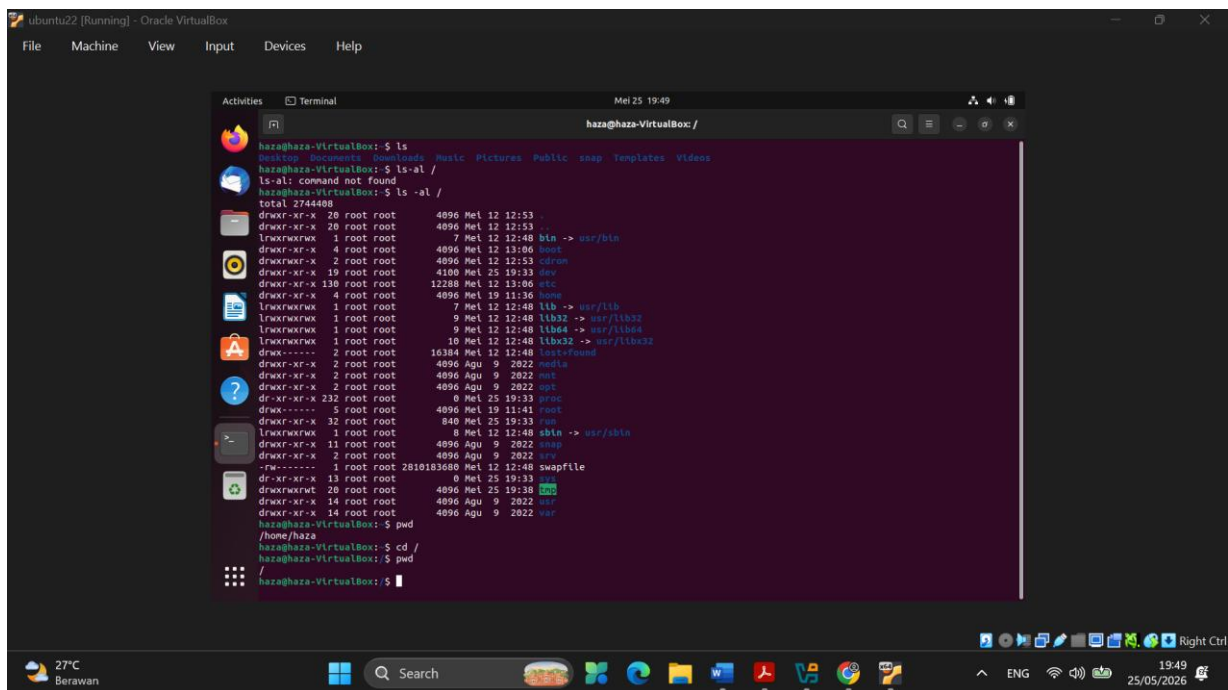
lalu Enter dan screenshot hasilnya.

Kalau mau memastikan sudah pindah ke root bisa lanjut:

`pwd`

hasilnya nanti:

`/`



The screenshot shows a terminal window titled 'haza@haza-VirtualBox: /'. The user has entered the command `cd /` and then `pwd`. The output of `pwd` is `/`, confirming the user is in the root directory. The terminal also shows the output of `ls -al /`, which lists the contents of the root directory, including `total 274448`, `drwxr-xr-x 28 root root 4096 Mei 12 12:53`, `drwxr-xr-x 20 root root 4096 Mei 12 12:53`, `lrwxrwxrwx 1 root root 7 Mei 12 12:48 bin -> usr/bin`, `drwxr-xr-x 4 root root 4096 Mei 12 11:06 dev`, `drwxrwxr-x 2 root root 4096 Mei 12 12:53 dram`, `drwxr-xr-x 19 root root 4100 Mei 25 19:33 dev`, `drwxr-xr-x 138 root root 12288 Mei 12 11:06 etc`, `drwxr-xr-x 4 root root 4096 Mei 19 11:36 home`, `lrwxrwxrwx 1 root root 7 Mei 12 12:48 lib -> usr/lib`, `lrwxrwxrwx 1 root root 9 Mei 12 12:48 lib32 -> usr/lib32`, `lrwxrwxrwx 1 root root 9 Mei 12 12:48 lib64 -> usr/lib64`, `lrwxrwxrwx 1 root root 10 Mei 12 12:48 libx32 -> usr/libx32`, `drwx----- 2 root root 16384 Mei 12 12:48 lost+found`, `drwxr-xr-x 2 root root 4096 Agu 9 2022 media`, `drwxr-xr-x 2 root root 4096 Agu 9 2022 net`, `drwxr-xr-x 2 root root 4096 Agu 9 2022 opt`, `dr-xr-xr-x 232 root root 0 Mei 25 19:33 proc`, `drwx----- 5 root root 4096 Mei 19 11:41 root`, `drwxr-xr-x 32 root root 840 Mei 25 19:33 run`, `lrwxrwxrwx 1 root root 8 Mei 12 12:48/sbin -> usr/sbin`, `drwxr-xr-x 11 root root 4096 Agu 9 2022 snap`, `drwxr-xr-x 2 root root 4096 Agu 9 2022 srv`, `-rwx----- 1 root root 2810183680 Mei 12 12:48 swapfile`, `dr-xr-xr-x 13 root root 0 Mei 25 19:33 sys`, `drwxrwxrwt 26 root root 4096 Mei 25 19:33 tmp`, `drwxr-xr-x 14 root root 4096 Agu 9 2022 usr`, and `drwxr-xr-x 14 root root 4096 Agu 9 2022 var`. The user then enters `pwd` and the output is `/`.

b. Option dan parameter dari perintah `cd /`

- option = tidak ada
- parameter = /

c. apa yg dilakukan perintah tsb

Perintah `cd /` digunakan untuk berpindah ke direktori root / pada Linux.

5. Direktori Khusus

a. Jalankan perintah

Ketik satu per satu:

```
cd /
```

lalu:

```
pwd
```

hasilnya:

```
/
```

Terus lanjut:

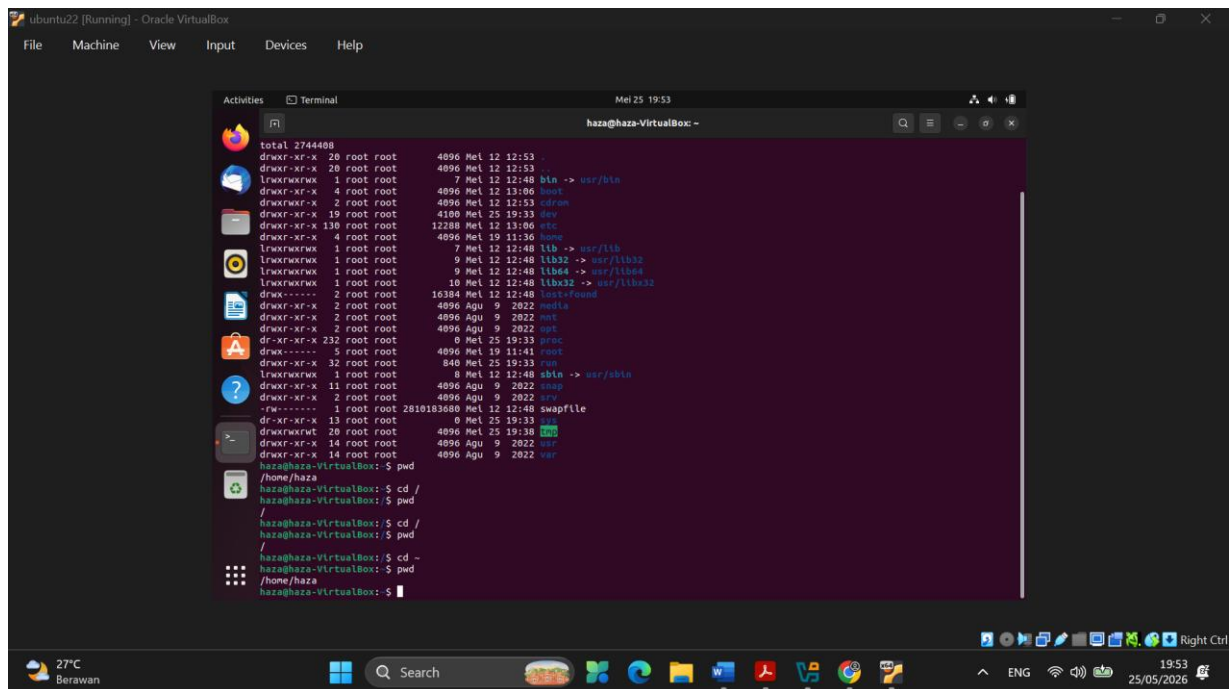
```
cd ~
```

cek lagi:

```
pwd
```

hasilnya biasanya:

```
/home/haza
```



Penjelasan hasil keduanya

- `cd /` → berpindah ke direktori root /
- `cd ~` → berpindah ke home directory user

Pada user kamu:

```
/home/haza
```

b. Jalankan perintah

Ketik:

```
cd /proc/self
```

cek lokasi:

```
pwd
```

hasil biasanya:

```
/proc/1234
```

atau angka lain.

Sekarang gunakan:

```
cd ..
```

ulang terus sambil cek:

```
pwd
```

Sampai hasilnya:

```
/
```

Jawaban

Perintah `cd ..` digunakan untuk kembali ke parent directory.

Biasanya perlu dijalankan:

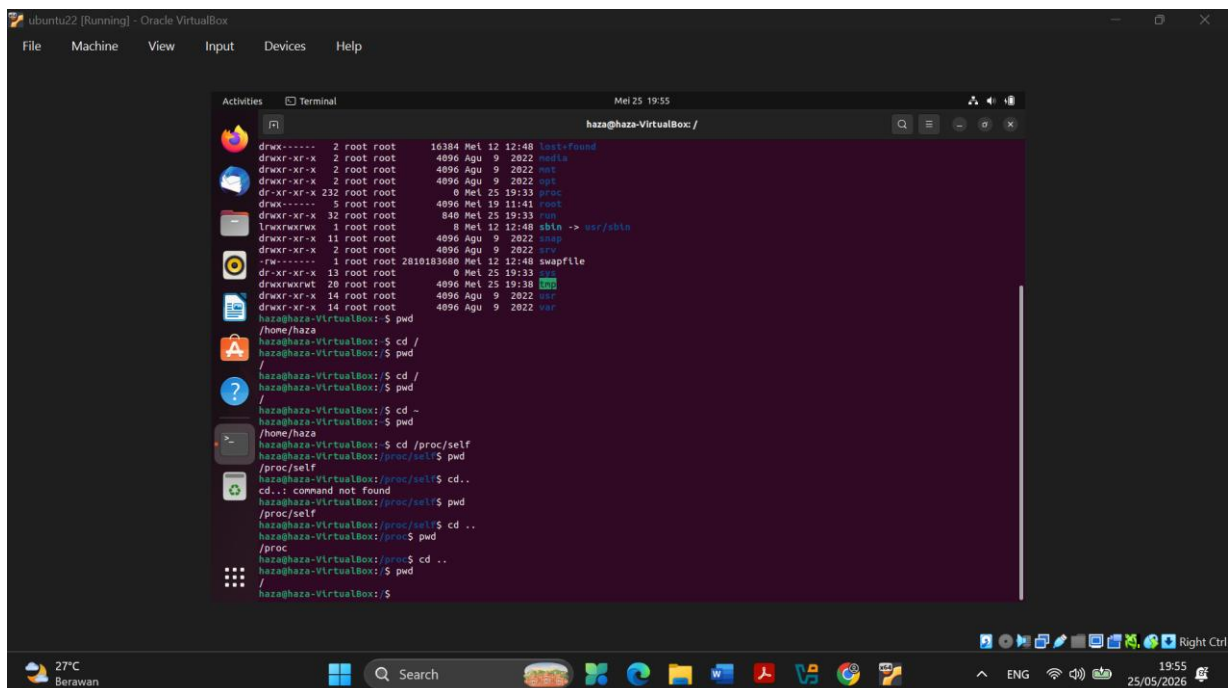
2 kali

Karena perpindahannya:

```
/proc/self
```

```
→ /proc
```

```
→ /
```



```
haza@haza-VirtualBox: /
drwxr-xr-x 2 root root 16384 Met 12 12:48 lost+found
drwxr-xr-x 2 root root 4096 Agu 9 2022 media
drwxr-xr-x 2 root root 4096 Agu 9 2022 mnt
drwxr-xr-x 232 root root 0 Met 25 19:33 /proc
drwxr-xr-x 5 root root 4096 Met 19 11:41 root
drwxr-xr-x 32 root root 840 Met 25 19:33 run
lrwxrwxrwx 1 root root 8 Met 12 12:48/sbin -> usr/sbin
drwxr-xr-x 11 root root 4096 Agu 9 2022 snap
drwxr-xr-x 2 root root 4096 Agu 9 2022 srv
-rw-r--r-- 1 root root 2010183680 Met 12 12:48 swapfile
dr-xr-xr-x 13 root root 0 Met 25 19:33 sys
drwxrwxrwt 26 root root 4096 Met 25 19:38 tmp
drwxr-xr-x 14 root root 4096 Agu 9 2022 var
haza@haza-VirtualBox: $ pwd
/home/haza
haza@haza-VirtualBox: $ cd /
haza@haza-VirtualBox: $ pwd
/
haza@haza-VirtualBox: $ cd /
haza@haza-VirtualBox: $ pwd
/
haza@haza-VirtualBox: $ cd ..
haza@haza-VirtualBox: $ pwd
/home/haza
haza@haza-VirtualBox: $ cd /proc/self
haza@haza-VirtualBox: /proc/self $ pwd
/proc/self
haza@haza-VirtualBox: /proc/self $ cd ..
cd.: command not found
haza@haza-VirtualBox: /proc/self $ pwd
/proc/self
haza@haza-VirtualBox: /proc/self $ cd ..
haza@haza-VirtualBox: /proc $ pwd
/proc
haza@haza-VirtualBox: /proc $ cd ..
haza@haza-VirtualBox: $ pwd
/
haza@haza-VirtualBox: $
```


6.

```

haza@haza-VirtualBox: /
haza@haza-VirtualBox:~$ cd /
haza@haza-VirtualBox:/$ pwd
/
haza@haza-VirtualBox:/$ cd /
haza@haza-VirtualBox:/$ pwd
/
haza@haza-VirtualBox:/$ cd -
haza@haza-VirtualBox:/$ pwd
/home/haza
haza@haza-VirtualBox:/$ cd /proc/self
haza@haza-VirtualBox:/proc/self$ pwd
/proc/self
haza@haza-VirtualBox:/proc/self$ cd..
cd..: command not found
haza@haza-VirtualBox:/proc/self$ pwd
/proc/self
haza@haza-VirtualBox:/proc/self$ cd ..
haza@haza-VirtualBox:/proc$ pwd
/proc
haza@haza-VirtualBox:/proc$ cd ..
haza@haza-VirtualBox:/$ pwd
/
haza@haza-VirtualBox:/$ cp /proc/cpuinfo ~/
haza@haza-VirtualBox:/$ ls -
cpuinfo Desktop Documents Downloads Music Pictures Public snap Templates Videos
haza@haza-VirtualBox:/$ cp /proc/uptime ~/
haza@haza-VirtualBox:/$ ls -
cpuinfo Desktop Documents Downloads Music Pictures Public snap Templates uptime Videos
haza@haza-VirtualBox:/$ rm ~/uptime
rm: remove write-protected regular file '/home/haza/uptime'?
haza@haza-VirtualBox:/$ ls -
cpuinfo Desktop Documents Downloads Music Pictures Public snap Templates uptime Videos
haza@haza-VirtualBox:/$ rm ~/uptime
rm: remove write-protected regular file '/home/haza/uptime'? Y
haza@haza-VirtualBox:/$ ls -
cpuinfo Desktop Documents Downloads infocpu Music Pictures Public snap Templates Videos
haza@haza-VirtualBox:/$ mv ~/cpuinfo ~/infocpu
haza@haza-VirtualBox:/$ ls -
Desktop Documents Downloads infocpu Music Pictures Public snap Templates Videos
haza@haza-VirtualBox:/$

```

Copy, Rename dan Delete File

Screenshot setiap langkah yaa

a. Copy file cpuinfo ke home

Ketik:

```
cp /proc/cpuinfo ~/
```

b. Tunjukkan file sudah tercopy

Ketik:

```
ls ~
```

Nanti harus muncul:

```
cpuinfo
```

c. Copy file uptime ke home

Ketik:

```
cp /proc/uptime ~/
```

d. Tunjukkan file sudah tercopy

Ketik:

```
ls ~
```

Nanti muncul:

```
cpuinfo
```

```
uptime
```

e. Hapus file uptime

Ketik:

```
rm ~/uptime
```

f. Tunjukkan file sudah terhapus

Ketik:

```
ls ~
```

Nanti uptime sudah hilang.

g. Rename cpuinfo menjadi infocpu

Ketik:

```
mv ~/cpuinfo ~/infocpu
```

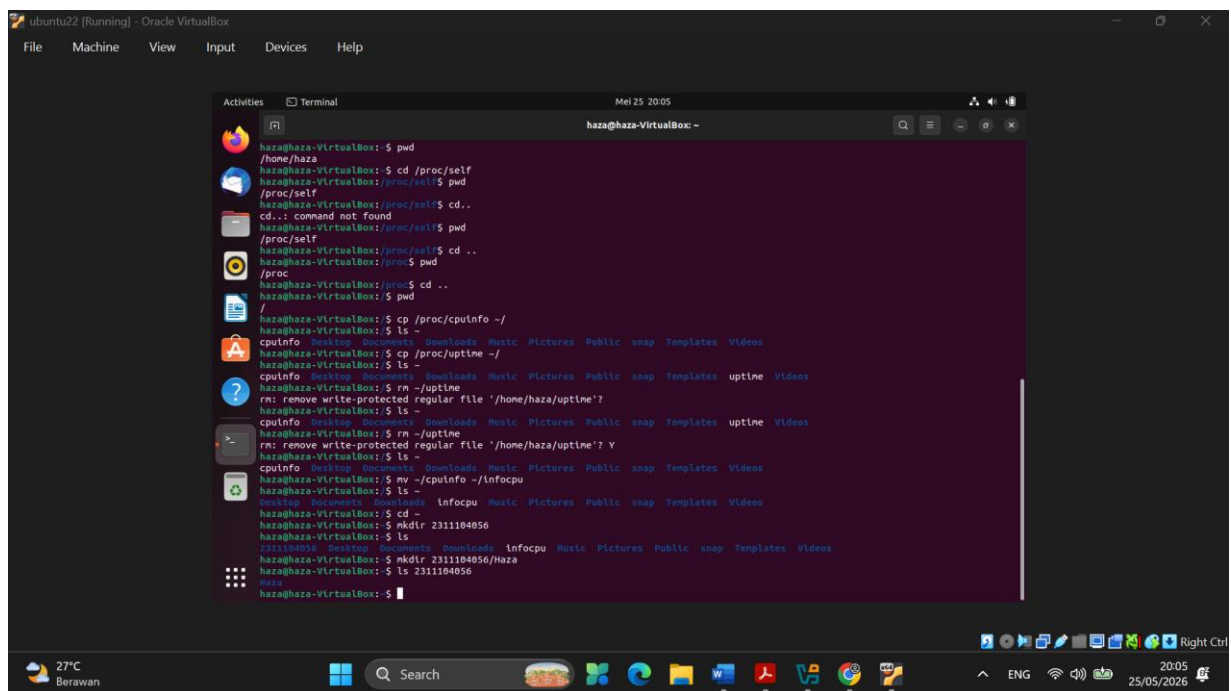
Cek:

```
ls ~
```

Nanti berubah jadi:

infocpu

7.



```
haza@haza-VirtualBox:~$ pwd
/home/haza
haza@haza-VirtualBox:~$ cd /proc/self
haza@haza-VirtualBox:/proc/self$ pwd
/proc/self
haza@haza-VirtualBox:/proc/self$ cd ..
cd.: command not found
haza@haza-VirtualBox:/proc/self$ pwd
/proc/self
haza@haza-VirtualBox:/proc/self$ cd ..
haza@haza-VirtualBox:/proc$ pwd
/proc
haza@haza-VirtualBox:/proc$ cd ..
haza@haza-VirtualBox:$ pwd
/
haza@haza-VirtualBox:$ cp /proc/cpuinfo ~/
haza@haza-VirtualBox:$ ls -
cpuinfo Desktop Documents Downloads Music Pictures Public snap Templates Videos
haza@haza-VirtualBox:$ cp /proc/uptime ~/
haza@haza-VirtualBox:$ ls -
cpuinfo Desktop Documents Downloads Music Pictures Public snap Templates uptime Videos
haza@haza-VirtualBox:$ rm ~/uptime
rm: remove write-protected regular file '/home/haza/uptime'?
haza@haza-VirtualBox:$ ls -
cpuinfo Desktop Documents Downloads Music Pictures Public snap Templates uptime Videos
haza@haza-VirtualBox:$ rm ~/uptime
rm: remove write-protected regular file '/home/haza/uptime'? Y
haza@haza-VirtualBox:$ mv ~/cpuinfo ~/infocpu
haza@haza-VirtualBox:$ cd -
~/Desktop/Downloads/infocpu
haza@haza-VirtualBox:$ mkdir 2311104056
haza@haza-VirtualBox:$ mkdir 2311104056/haza
haza@haza-VirtualBox:$ ls 2311104056
haza
```

7. Membuat Folder Baru

Awali dulu dengan:

```
cd ~
```

a. Buat folder dengan nama NIM

Contoh:

```
mkdir 231011400123
```

Ganti dengan NIM kamu sendiri

Cek:

```
ls
```

nanti folder NIM muncul.

b. Buat folder nama di dalam folder NIM

Contoh:

```
mkdir 231011400123/Haza
```

Ganti Haza dengan nama kamu.

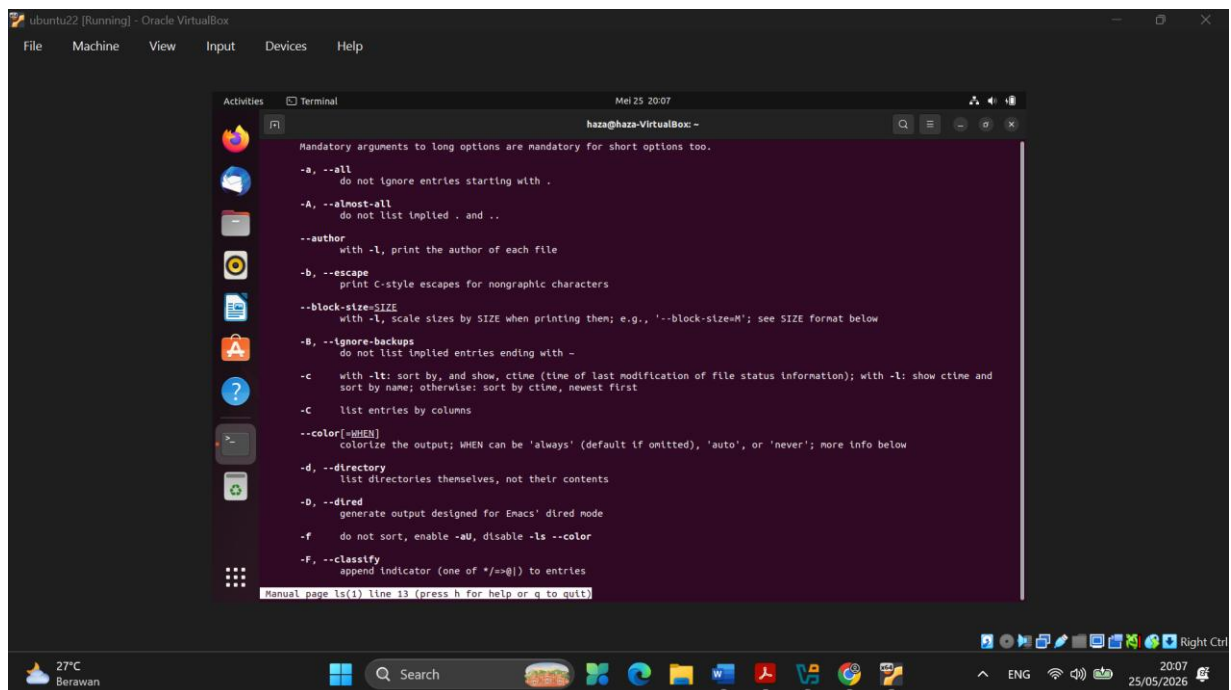
Cek:

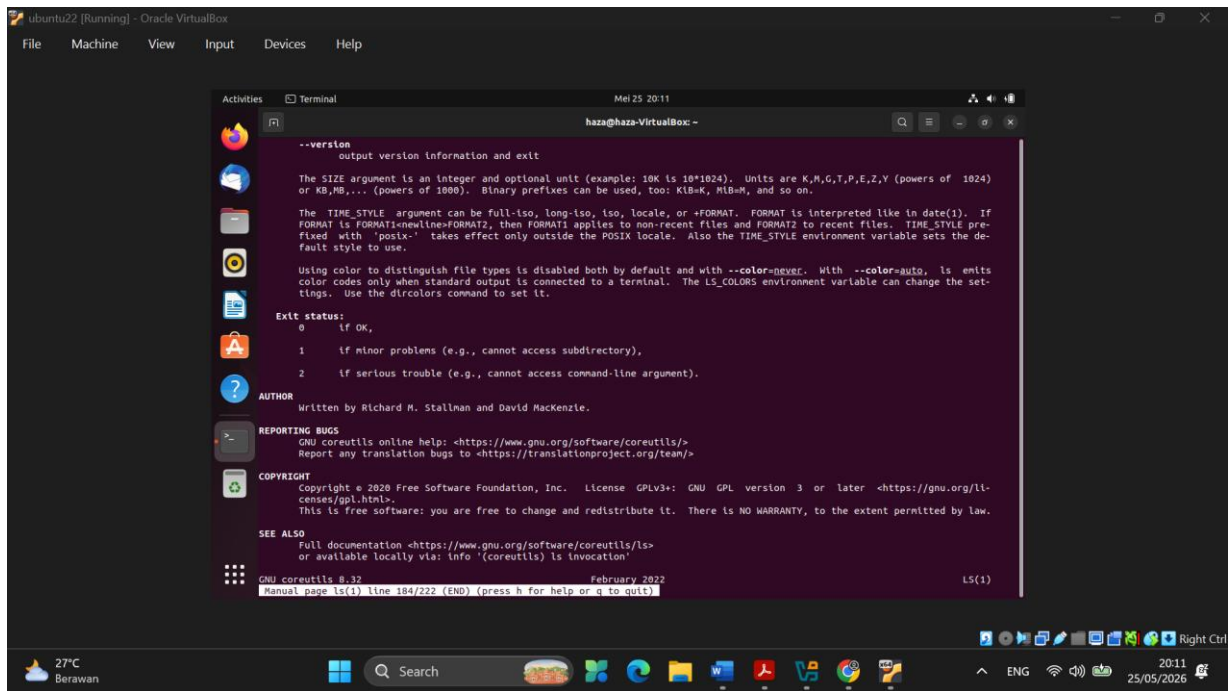
```
ls 231011400123
```

nanti muncul:

Haza

8.





Membaca Manual

a. Buka manual ls

Ketik:

man ls

Untuk keluar dari manual tekan:

q

Screenshot halaman manualnya

b. Fungsi perintah ls

Fungsi perintah ls adalah menampilkan isi direktori/folder.

c. Siapa pencipta ls

Biasanya tertulis di manual:

Richard M. Stallman dan David MacKenzie

d. Arti -h pada manual ls

-h

artinya:

human readable

Ukuran file ditampilkan lebih mudah dibaca seperti KB, MB, GB.

e. Option untuk melihat direktori secara rekursif

-R

f. Buka manual cp

Ketik:

man cp

keluar dengan:

q

g. Fungsi perintah cp

Untuk menyalin/copy file atau folder.

h. Siapa pencipta cp

Biasanya tertulis:

Torbjorn Granlund, David MacKenzie, dan Jim Meyering

i. Arti -v pada cp

-v

artinya:

verbose

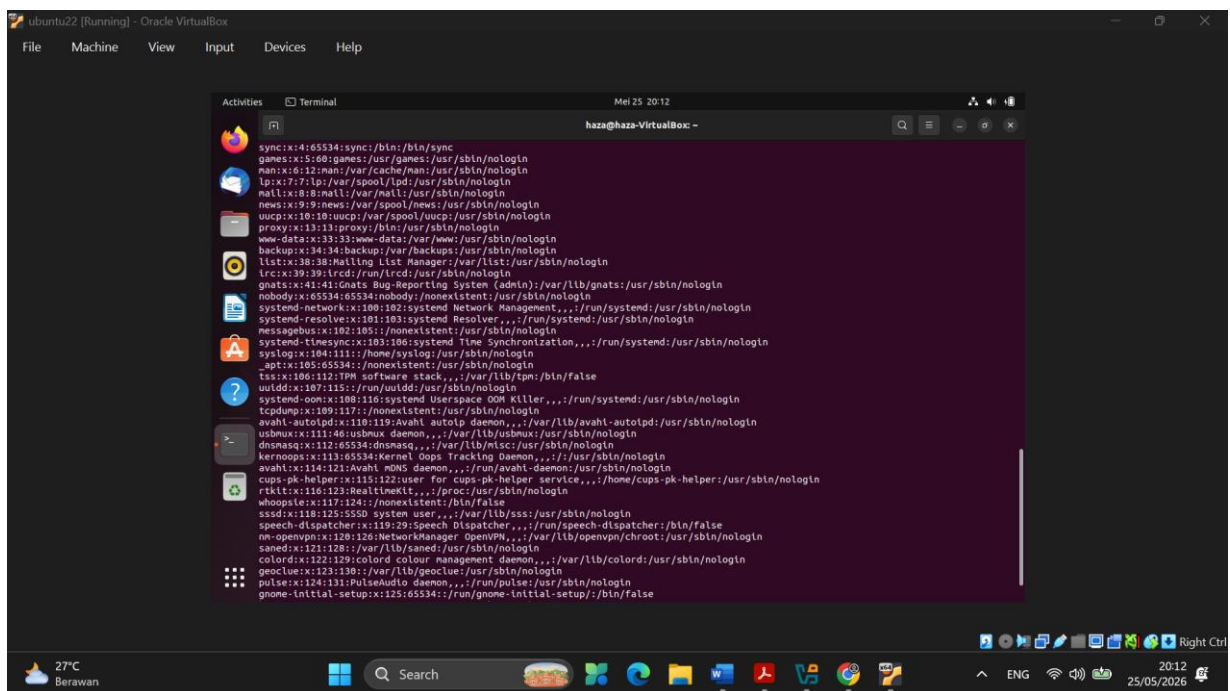
Menampilkan proses copy yang dilakukan.

j. Option agar interaktif

-i

Digunakan agar muncul konfirmasi sebelum overwrite file.

9.



a. Jalankan perintah

Ketik:

cat /etc/passwd

lalu screenshot hasilnya.

b. Fungsi perintah cat

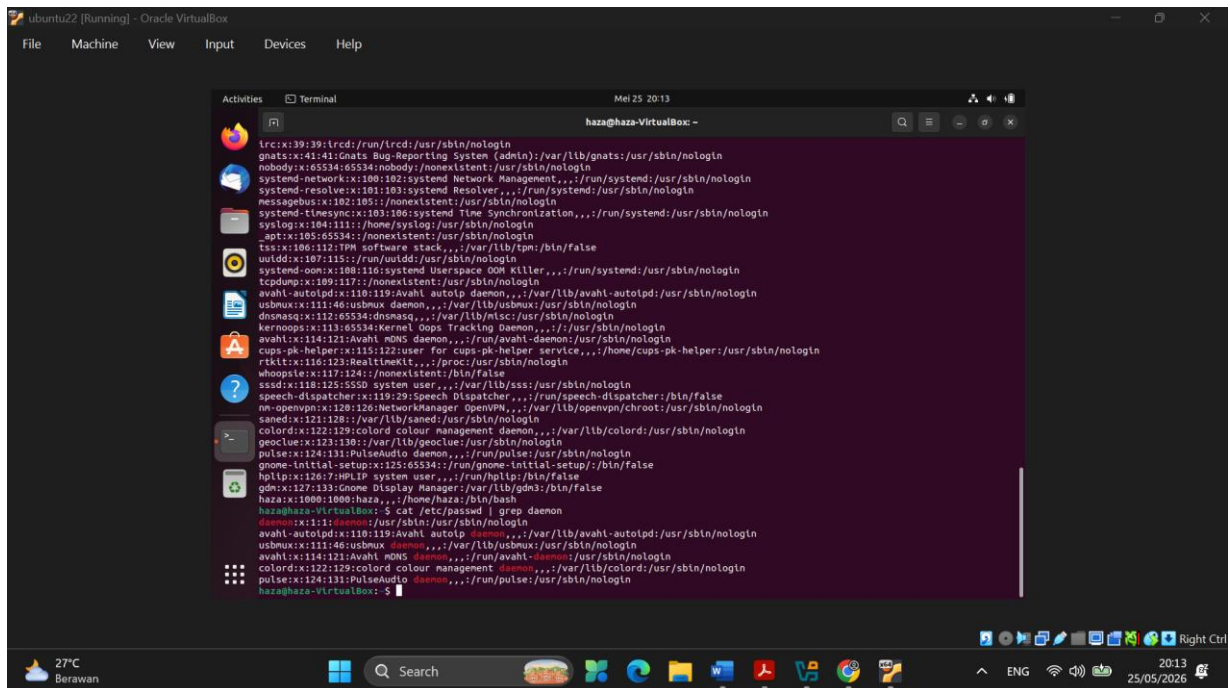
Fungsi cat adalah untuk menampilkan isi file.

c. Jalankan perintah

Ketik:

cat /etc/passwd | grep daemon

Screenshot hasilnya.

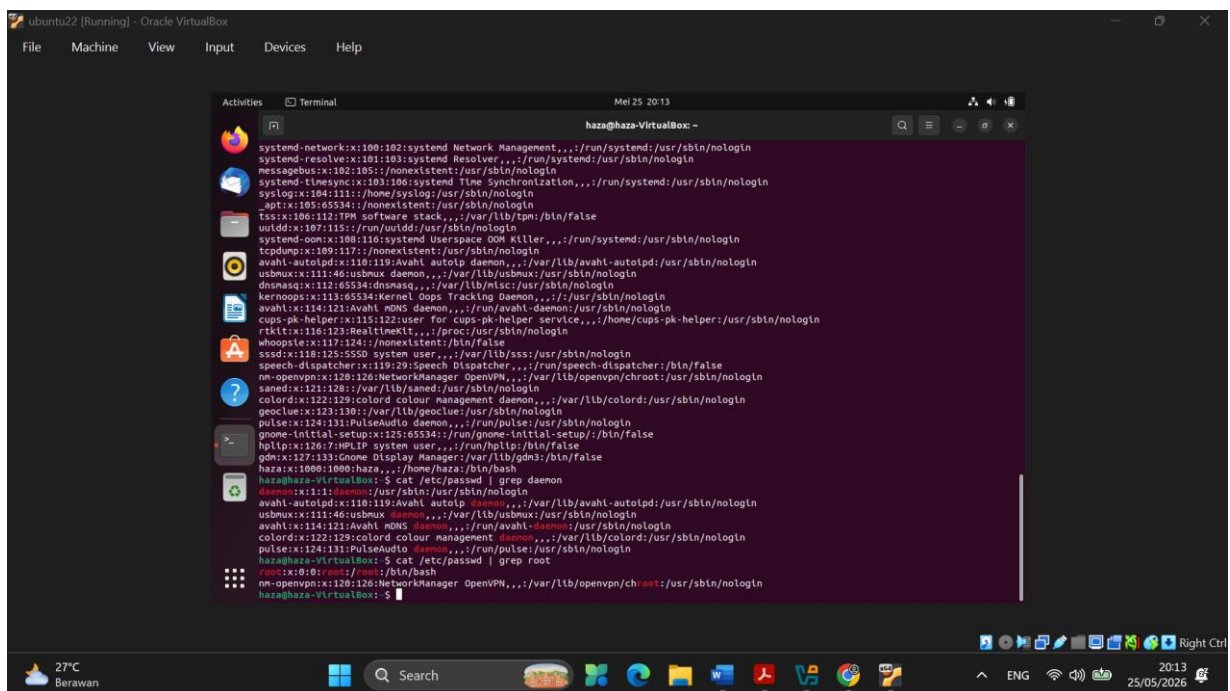


```
haza@haza-VirtualBox: ~  
lrc:x:39:39:ircd:/run/lrcd:/usr/sbin/nologin  
gnats:x:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/usr/sbin/nologin  
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin  
systemd-network:x:100:102:systemd Network Management,,,:/run/systemd:/usr/sbin/nologin  
systemd-resolve:x:101:103:systemd Resolver,,,:/run/systemd:/usr/sbin/nologin  
messagebus:x:102:105:102:105:/nonexistent:/usr/sbin/nologin  
systemd-timesync:x:103:106:systemd Time Synchronization,,,:/run/systemd:/usr/sbin/nologin  
syslog:x:104:111:104:111:/home/syslog:/usr/sbin/nologin  
_apt:x:105:65534:105:65534:/nonexistent:/usr/sbin/nologin  
tss:x:106:112:TPM software stack,,,:/var/lib/tpm:/bin/false  
uuidd:x:107:115:107:115:/run/uuidd:/usr/sbin/nologin  
systemd-oom:x:108:116:systemd Userspace OOM Killer,,,:/run/systemd:/usr/sbin/nologin  
tcpdump:x:109:117:109:117:/nonexistent:/usr/sbin/nologin  
avahi-autopd:x:110:119:Avahi autopd daemon,,,:/var/lib/avahi-autopd:/usr/sbin/nologin  
usbmux:x:111:46:usbmux daemon,,,:/var/lib/usbmux:/usr/sbin/nologin  
dnsmasq:x:112:65534:dnsmasq,,,:/var/lib/mtsc:/usr/sbin/nologin  
kernoops:x:113:65534:Kernel Oops Tracking Daemon,,,:/usr/sbin/nologin  
avahi:x:114:121:Avahi mDNS daemon,,,:/run/avahi-daemon:/usr/sbin/nologin  
cups-pk-helper:x:115:122:user for cups-pk-helper service,,,:/home/cups-pk-helper:/usr/sbin/nologin  
rtkit:x:116:123:RealtimeKit,,,:/proc:/usr/sbin/nologin  
whoopsie:x:117:124:117:124:/nonexistent:/bin/false  
sssd:x:118:125:SSSD system user,,,:/var/lib/sss:/usr/sbin/nologin  
speech-dispatcher:x:119:29:Speech Dispatcher,,,:/run/speech-dispatcher:/bin/false  
nn-opensvn:x:120:126:NetworkManager OpenVPN,,,:/var/lib/opensvn/chronot:/usr/sbin/nologin  
saned:x:121:128:121:128:/var/lib/saned:/usr/sbin/nologin  
colord:x:122:129:colord colour management daemon,,,:/var/lib/colord:/usr/sbin/nologin  
geoclue:x:123:130:123:130:/var/lib/geoclue:/usr/sbin/nologin  
pulse:x:124:131:PulseAudio daemon,,,:/run/pulse:/usr/sbin/nologin  
gnome-initial-setup:x:125:65534:125:65534:/run/gnome-initial-setup:/bin/false  
hplip:x:126:7:HPLIP system user,,,:/run/hplip:/bin/false  
gdm:x:127:133:Gnome Display Manager:/var/lib/gdm3:/bin/false  
haza:x:1000:1000:haza,,,:/home/haza:/bin/bash  
haza@haza-VirtualBox: ~$ cat /etc/passwd | grep daemon  
daemon:x:111:daemon:/usr/sbin:/usr/sbin/nologin  
avahi-autopd:x:110:119:Avahi autopd daemon,,,:/var/lib/avahi-autopd:/usr/sbin/nologin  
usbmux:x:111:46:usbmux daemon,,,:/var/lib/usbmux:/usr/sbin/nologin  
avahi:x:114:121:Avahi mDNS daemon,,,:/run/avahi-daemon:/usr/sbin/nologin  
colord:x:122:129:colord colour management daemon,,,:/var/lib/colord:/usr/sbin/nologin  
pulse:x:124:131:PulseAudio daemon,,,:/run/pulse:/usr/sbin/nologin  
haza@haza-VirtualBox: ~$
```

d. Jalankan perintah

cat /etc/passwd | grep root

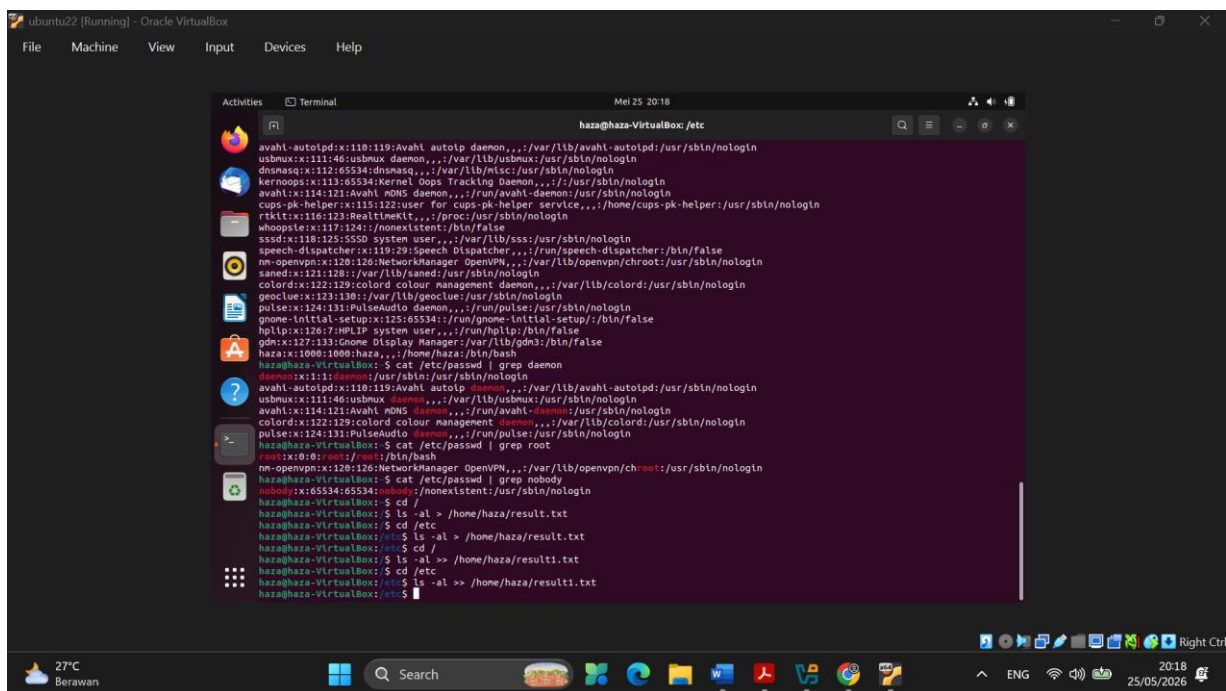
Screenshot hasilnya.



```
haza@haza-VirtualBox: ~  
systemd-network:x:100:102:systemd Network Management,,,:/run/systemd:/usr/sbin/nologin  
systemd-resolve:x:101:103:systemd Resolver,,,:/run/systemd:/usr/sbin/nologin  
messagebus:x:102:105:102:105:/nonexistent:/usr/sbin/nologin  
systemd-timesync:x:103:106:systemd Time Synchronization,,,:/run/systemd:/usr/sbin/nologin  
_apt:x:105:65534:105:65534:/nonexistent:/usr/sbin/nologin  
tss:x:106:112:TPM software stack,,,:/var/lib/tpm:/bin/false  
uuidd:x:107:115:107:115:/run/uuidd:/usr/sbin/nologin  
systemd-oom:x:108:116:systemd Userspace OOM Killer,,,:/run/systemd:/usr/sbin/nologin  
tcpdump:x:109:117:109:117:/nonexistent:/usr/sbin/nologin  
avahi-autopd:x:110:119:Avahi autopd daemon,,,:/var/lib/avahi-autopd:/usr/sbin/nologin  
usbmux:x:111:46:usbmux daemon,,,:/var/lib/usbmux:/usr/sbin/nologin  
dnsmasq:x:112:65534:dnsmasq,,,:/var/lib/mtsc:/usr/sbin/nologin  
kernoops:x:113:65534:Kernel Oops Tracking Daemon,,,:/usr/sbin/nologin  
avahi:x:114:121:Avahi mDNS daemon,,,:/run/avahi-daemon:/usr/sbin/nologin  
cups-pk-helper:x:115:122:user for cups-pk-helper service,,,:/home/cups-pk-helper:/usr/sbin/nologin  
rtkit:x:116:123:RealtimeKit,,,:/proc:/usr/sbin/nologin  
whoopsie:x:117:124:117:124:/nonexistent:/bin/false  
sssd:x:118:125:SSSD system user,,,:/var/lib/sss:/usr/sbin/nologin  
speech-dispatcher:x:119:29:Speech Dispatcher,,,:/run/speech-dispatcher:/bin/false  
nn-opensvn:x:120:126:NetworkManager OpenVPN,,,:/var/lib/opensvn/chronot:/usr/sbin/nologin  
saned:x:121:128:121:128:/var/lib/saned:/usr/sbin/nologin  
colord:x:122:129:colord colour management daemon,,,:/var/lib/colord:/usr/sbin/nologin  
geoclue:x:123:130:123:130:/var/lib/geoclue:/usr/sbin/nologin  
pulse:x:124:131:PulseAudio daemon,,,:/run/pulse:/usr/sbin/nologin  
gnome-initial-setup:x:125:65534:125:65534:/run/gnome-initial-setup:/bin/false  
hplip:x:126:7:HPLIP system user,,,:/run/hplip:/bin/false  
gdm:x:127:133:Gnome Display Manager:/var/lib/gdm3:/bin/false  
haza:x:1000:1000:haza,,,:/home/haza:/bin/bash  
haza@haza-VirtualBox: ~$ cat /etc/passwd | grep root  
root:x:0:0:root:/root:/bin/bash  
nn-opensvn:x:120:126:NetworkManager OpenVPN,,,:/var/lib/opensvn/chronot:/usr/sbin/nologin  
haza@haza-VirtualBox: ~$
```


- | = pipe (menghubungkan output ke perintah lain)
- grep daemon = mencari kata daemon pada hasil output sebelumnya.

10.



a. Jalankan perintah

Ketik:

cd /

ls -al > /home/haza/result.txt

Screenshot terminalnya.

Penjelasan hasil

Perintah tersebut:

- masuk ke direktori root /
- menjalankan ls -al
- hasil output disimpan ke file result.txt

Output tidak tampil di terminal karena diarahkan ke file.

b. Dimana file result.txt berada?

File berada di:

/home/haza/result.txt

c. Jalankan perintah

Ketik:

cd /etc

ls -al > /home/haza/result.txt

Penjelasan hasil

Isi file result.txt sebelumnya akan diganti/ditimpa dengan hasil baru dari direktori /etc.

d. Fungsi >

Fungsi > adalah:

mengalihkan output ke file dan menimpa isi file lama

e. Jalankan perintah

Ketik:

```
cd /
```

```
ls -al >> /home/haza/result1.txt
```

f. Jalankan perintah

Ketik:

```
cd /etc
```

```
ls -al >> /home/haza/result1.txt
```

Penjelasan hasil

Hasil perintah kedua akan ditambahkan ke dalam file result1.txt tanpa menghapus isi sebelumnya.

g. Perbedaan > dan >>

- > = overwrite / menimpa isi file
- >> = append / menambahkan isi ke file tanpa menghapus isi lama